ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3 m s^{-1}	120 ÷ 40
2	**(d)** 10 m s^{-1}	36 × 5/18
3	**(d)** 5 m s^{-2}	(30 − 10) ÷ 4
4	**(d)** 21 m s^{-1}	v = u + at
5	**(c)** 18 m	s = ut + ½at^2
6	**(c)** 44 m and 0	circumference = 2πr; returns to start
7	**(a)** m s^{-2}	change in velocity per second
8	**(a)** displacement	v × t
9	**(d)** constant speed but changing velocity	direction keeps changing
10	**(a)** acceleration is opposite to the chosen positive direction	sign gives direction

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance is the total path length (scalar, only magnitude); displacement is the net change in position with direction (1+1).	1+1
12	No; the straight-line change in position cannot exceed the actual path length (2).	2
13	Displacement divided by the time interval (1); m s^{-1} (1).	1+1
14	Distance = 3 + 4 = 7 m (1); displacement = √(3^2 + 4^2) = 5 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A girl runs 100 m east and returns 40 m (1); distance = 140 m (1); displacement = 60 m east (1).	1+1+1
16	Distance = 12 + 6 = 18 m (2); displacement = 6 m upward (1).	2+1
17	Average speed = 400/5 = 80 km h^{-1} (2); average velocity = 0 (1).	2+1
	OR: When it returns to the starting point, displacement is zero but distance is not (3).	3
18	36 km/h = 10 m/s, 54 km/h = 15 m/s (1); a = (15 − 10)/10 (1) = 0.5 m s^{-2} (1).	1+1+1
	OR: u = 15 m/s; a = −15/5 = −3 m s^{-2} (1.5); s = 15 × 5/2 = 37.5 m (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Net change in position; displacement/time (2). (ii) Distance 160 m; displacement 40 m (2). (iii) Speed = distance/time (scalar); velocity = displacement/time (vector) (1).	2+2+1
	OR: (i) Rate of change of velocity; m s^{-2} (1). (ii) 10 → 20 m/s; a = 2 m s^{-2} (2). (iii) Positive: along velocity (speeding up), negative: opposite (slowing) (2).	1+2+2
20	(i) 10 and 15 m/s (1). (ii) 0.5 m s^{-2} (1). (iii) −3 m s^{-2} (1). (iv) 15 × 5/2 = 37.5 m (2).	1+1+1+2
	OR: (i) Time on x-axis, position on y-axis, scale, plot points and join (2). (ii) Velocity (1). (iii) Straight line vs curve (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5 m s^{-1}	300 ÷ 60
2	**(c)** 20 m s^{-1}	72 × 5/18
3	**(c)** −2 m s^{-2}	(5 − 25) ÷ 10
4	**(b)** 8 m s^{-1}	v = u + at
5	**(b)** 40 m	$v^2 = u^2 + 2as$
6	**(b)** 88 m and 0	circumference = 2πr; returns to start
7	**(c)** moves in a straight line in one direction	distance = |displacement|
8	**(c)** at rest	position does not change
9	**(b)** along the tangent	tangent to circle
10	**(b)** 100 m	average velocity 10 m/s × 10 s

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance = 2 × 22/7 × 7 = 44 m (1); speed = 44/40 = 1.1 m s^{-1} (1).	1+1
12	90 × 5/18 = 25 m s^{-1} (2).	2
13	Rate of change of velocity (1); acceleration opposite to velocity, i.e. retardation if it is slowing down (1).	1+1
14	a = (20 − 10)/5 (1) = 2 m s^{-2} (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) Straight line parallel to the time axis (1.5); (ii) straight inclined line whose slope gives constant velocity (1.5).	1½+1½
16	Uniform: straight line (1.5); non-uniform: curve, since displacements in equal intervals are unequal (1.5).	1½+1½
17	Straight line inclined to the time axis (1); take two points, acceleration = $(v_2 - v_1)/(t_2 - t_1)$ = slope (2).	1+2
	OR: Slope zero, so acceleration = 0 (1); displacement = area = 20 × 6 = 120 m (2).	1+2
18	Area of trapezium = ½ × (5 + 15) × 10 = 100 m (3).	3
	OR: a = (v − u)/t (2); hence v = u + at (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Horizontal line; inclined line (2). (ii) Slope (1). (iii) ½ × (5 + 15) × 10 = 100 m (2).	2+1+2
	OR: (i) Velocity increases by equal amounts in equal times — uniform acceleration (1). (ii) 10/20 = 0.5 m s^{-2} (1). (iii) ½ × 20 × 10 = 100 m (2). (iv) Straight line through origin (1).	1+1+2+1
20	(i) From a = (v − u)/t (1). (ii) Area of trapezium/rectangle plus triangle (2). (iii) Eliminate t: t = (v − u)/a; s = (v + u)(v − u)/2a (2).	1+2+2
	OR: (i) 20 m/s (1). (ii) 100 m (1). (iii) 400/10 = 40 m (2). (iv) 140 m (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.